Clara Lage

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Current Position

Since 2022 **Postdoctoral researcher in Signal Processing**, *ENS Lyon*, (Physics Lab), Main Topics: Optimization, Signal Process, Biology.
Supervisors: Nelly Pustelnik, Benjamin Audit

Postdoc

2020 – 2022 **Postdoctoral researcher in Applied Mathematics**, *Ecole Polytechnique (Paris)*, (CMAP), Stress Testing chair: BNP Parisbas. Main Topics: Optimization, Optimal Transport, Credit Portfolio, Climate Finance.
Supervisor: Emanuel Gobet

Education

2017 – 2020 **Ph.D Student in Mathematics**, *Panthéon-Sorbonne University and Instituto Nacional De Matemática Pura e Aplicada (IMPA) in a CIFRE thesis with ENGIE.*, Main Topics: Stochastic Optimization, Bilevel optimization, Price of Hydro-Energy, Dual Regularization.
Advisor: Mikhail Solodov
Advisor: Jean-Marc Bonniseau
Co-advisor: Claudia Sagastizábal
Scholarship from: Association Nationale Recherche Technologie (ANRT) and Conselho Nacional de Desenvolvimento Cientifico e Tecnologico (CNPQ)

2013–2014 **Mathematics M.S.**, *Instituto Nacional de Matemática Pura e Aplicada (IMPA)*, Rio de Janeiro.
Advisor: Felipe Linares
Specialization in Probability (Markov Chains and Stochastic Process)
Scholarship from: Conselho Nacional de Desenvolvimento Científico Tecnológico (CNPQ)

2009–2012 **Mathematics B.S**, *Universidade Federal de Santa Catarina (UFSC)*, Florianópolis.

Awards

2024 **Dodu Award**, *Gaspard Monge Program for optimization (PGMO)*, Best young researcher presentation during the SMAI-MODE Days.

2012 **Honors in Universitary Brazilian National Olympiad**, *Sociedade Brasileira de Matemática (OBM)*.

Research Interests

Nonconvex optimization

Stochastic optimization

Signal processing

Biology applications

Energy applications

Publications

2025 Lage, Clara. Pustelnik, Nelly. Arbona, Jean-Michel. Audit, Benjamin. Gribonval, Rémi. Identifying a piecewise affine signal from its nonlinear observation - application to DNA replication analysis. *IEEE Transactions of Signal Processing* 99 (2025), 1–15.

2024 Lage, Clara. Pustelnik, Nelly. Arbona, Jean-Michel. Audit, Benjamin . Time-Scale Hybrid Continuous-Discrete Sliding Frank-Wolfe Method, *IEEE Signal Processing Letters 31 (2024), 830–834.*

2024 Structured Dictionary Learning of Rating Migration Matrices for Credit Risk Modeling. *Computational Statistics* 39 (2024), 3431–3456.

2023 Lage, Clara. Pustelnik, Nelly. Audit, Benjamin. Arbona, Jean-Michel. Codage espace-échelle parcimonieux en présence de bruit non-gaussien. Application à l'analyse de la réplication de l'ADN en molécule unique. Gretsi 2023.

2023 Lage, Clara. Gobet, Emmanuel. Optimal ecological transition path of a credit portfolio distribution, based on Multidate Monge-Kantorovich formulation. *Annals of Operation Reseach 336 (2023),1-35.*

2022 Lage, Clara. Sagastizabal, Claudia. Solodov, Mikhail. Erbs, Guillaume. Increasing Reliability of Price Signals in Long Term Energy Management Problems. *Computational Optimization and Applications* 85 (2023), 787–820.

2019 Lage, Clara. Sagastizabal, Claudia. Solodov, Mikhail. Multiplier Stabilization Applied to Two-Stage Stochastic Programs. *Journal of Optimization Theory and Applications.* 183.(1) (2019), 158–178.

Teaching Experience

Optimization: Theory and applications — Level: M2, Master: *Mathématiques Financières : Probabilités et Finance* at Ecole Polytechnique and Université Paris Dauphine. 25 hours course. Years: 2020 and 2021. Lecutre course (Cours magistral). Course delivered in French. (MAP652A)

Probability Theory — Level: L3 at Ecole Polytechnique. 19 hours course. Years: 2020 and 2021. Tutorial course (Travaux dirigés). Course delivered in French. (MAP361)

Statistics — Level: M1, Master: *Data Science for Business* at Ecole Polytechnique and HEC. 17 hours course. Years: 2020 and 2021. Tutorial course (Travaux dirigés). Course delivered in english. (MAP531)

Languages

Portuguese	**Native Speaker**
English	**Advanced**
French	**Advanced**

Computer skills

Currently programming in Python [5 years], Matlab [2 years] and GAMS [1 year], R [1 yeat], beginner in C++

Talks in Academic Events

2024 **Toulouse**, AISSAI - Heterogenous Data and Large Representation Models in Science.

2024 **Lyon**, SMAI-MODE - French Society of Applied and Industrial Mathematics (SMAI).

2023 **Grenoble**, Gretsi - Groupe de Recherche et d'Etudes de Traitement du Signal et des Images .

2022 **Paris**, Financial Forum: Portfolio Optimization.

2021 **Online Conference**, NeurIPS Workshop: Optimal Transport in Machine Learning.

2019 **Throndheim**, *ICSP*, International Conference on Stochastic Programming.

2016 **Rio de Janeiro**, *Stochastic Variational Analysis (SVAN)*.

2016 **Federal University of Amazonas (UFAM)**, *XI Brazilian Workshop in Continuous Optimization*.

Scientific Animation

2023 **Lyon**, Supervision of observation internship for middle school students.

2024 **Lyon**, Supervision of observation internship for ENS de Lyon students.

2024 **Lyon**, Organization of the Physics Laboratory Workshop at the Fête de la Science.